

3.1 General properties of waves

5054 Paper 2 only. 2015 onwards. Mark-carrying lines.

Definitions

Waves transfer energy without transferring matter.

Frequency: the number of wavelengths / waves / cycles / oscillations that pass a point per unit time. Also the number generated by the source per unit time. Unit Hz. $f = 1/T$.

Wavelength: the distance between two consecutive identical points, e.g. consecutive crests.

Amplitude: the maximum displacement / distance from the mean (undisturbed) position.

Wavefront: a line joining points that are in phase / at the same point in the motion / a line along a crest.

Compression: particles of the medium closer together; higher density and pressure.

Rarefaction: particles further apart; lower density and pressure.

One wavelength of a longitudinal wave is compression to next compression (or rarefaction to next rarefaction).

Compression to the next rarefaction is $\frac{1}{2}\lambda$.

$$v = f\lambda$$

Write the equation, convert to SI, substitute, give the unit.

Frequency from a source: number of oscillations or crests divided by the time. Per minute, divide by 60. From a period: $f = 1/T$.

Frequency stays the same when a wave changes medium or depth.

If speed decreases, wavelength decreases. If speed increases, wavelength increases.

If only frequency of the bar is increased in a ripple tank of fixed depth, speed stays the same and wavelength decreases.

Graphs and drawings

Amplitude is the vertical distance from the mean to a crest. Peak-to-trough is $2A$.

On a height–time graph, period is the time for one complete wave. Equal spacing of peaks means constant frequency.

A height–time graph that starts at the mean has the first crest at $T/4$.

On a scale drawing of crests, measure several gaps and convert using the stated scale.

Transverse and longitudinal

Transverse: vibration at right angles to the direction of energy transfer.

Examples: light, X-rays and other electromagnetic waves, waves on the surface of water, a wave on a rope or string, seismic S-waves.

Longitudinal: vibration parallel to the direction of energy transfer.

Examples: sound, ultrasound, seismic P-waves.

A wave on a string: particles of the string move at right angles to the travel of the wave.

Ultrasound is a longitudinal wave of higher frequency than audible sound. It needs a medium.

Reflection

At a plane surface, angle of incidence = angle of reflection.

Frequency, speed and wavelength stay the same.

Ripple tank: plane barrier; reflected wavefronts have the same spacing as the incident wavefronts.

Sound-reflection experiment: loudest signal when $i = r$.

Refraction

Refraction is due to a change of speed.

Water waves travel more slowly in shallow water than in deep water.

Deep to shallow: frequency the same, speed down, wavelength down, wavefronts closer together, direction bends towards the normal.

Shallow to deep: speed up, wavelength up, direction bends away from the normal.

Wavefronts stay continuous across the boundary.

One side of a wavefront reaches the boundary first, slows first, and the wavefront changes direction.

Ripple tank: glass sheet / block on the floor to make a shallow region.

Whole tank made shallower, no other change: speed down, wavelength down, frequency the same.

Diffraction

Diffraction is spreading of a wave through a gap or past an edge.

Greatest through a gap when the gap is smaller than / similar to the wavelength: larger wavelength, smaller gap.

Drawing through a gap smaller than λ : semicircular / circular wavefronts of the same wavelength, centred on the gap.

At an edge: wavefronts curl into the shadow as quarter-circles centred on the corner. Same wavelength.

Smaller wavelength: less diffraction at the edge.

Hearing through an open door to a person in the shadow: diffraction. Draw curved wavefronts that reach that person.

Ripple tank

Tank with water. Wooden bar or dipper on the surface, moved up and down (motor / rubber bands) to produce the wave. Lamp above and screen below, or a stroboscope, to see the crests.

Reflection: plane barrier.

Refraction: glass plate on the floor.

Diffraction: gap in a barrier, or a single edge of a block.

Energy transfer: a floating object / cork oscillates (up and down) and does not travel with the wave.

To increase frequency: move the bar up and down faster (or tighter rubber bands / lighter bar). Wavelength then decreases. Speed stays the same.

Written marks

State the definition in the syllabus words.

$v = f\lambda$, SI substitution, unit.

Transverse vs longitudinal: state both directions of vibration relative to energy transfer.

Wavefront drawing: continuous fronts, correct spacing, enough fronts (at least three where asked).

Ripple tank: how the wave is made and how it is seen.